

1. Smart Healthcare Management System (SDG 3: Good Health and Well-being)

A government hospital plans to implement a Smart Healthcare Management System to improve patient care and reduce waiting time. Patients should be able to register online, schedule appointments, access electronic health records, receive digital prescriptions, and make online payments. Doctors, nurses, pharmacists, and laboratory technicians interact with the system to update medical information. The administration also requires automatic emergency alerts, medicine inventory monitoring, and health analytics dashboards. During system analysis, it is observed that patient information is duplicated across multiple departments, leading to inconsistencies and delayed treatment.

Analyze the functional and non-functional requirements of the system. Identify the actors, classes, relationships, and interactions involved. Recommend appropriate UML diagrams to represent the system architecture and justify your choice. Suggest object-oriented design improvements that eliminate redundancy and improve healthcare service delivery while supporting SDG 3.

